

Built a **SOC-focused Linux security** monitoring project using **Splunk SIEM** to detect and analyze an attack lifecycle by ingesting Linux authentication, firewall, kernel, and system logs ([/var/log/auth.log](#), [/var/log/kern.log](#), [/var/log/syslog](#)). Network reconnaissance was identified through UFW firewall block events, followed by SSH brute-force attempts and a successful SSH login. Dashboards were created to visualize port scanning activity, failed SSH logins, and successful SSH access, and alerts were implemented to detect suspicious firewall and authentication behavior, demonstrating how SOC teams correlate host-based logs in Splunk SIEM to identify attacker activity across multiple stages.

Tools & Components Used (Detailed)

Splunk Enterprise (SIEM)

- Used as the **central log collection and analysis platform**
- Configured to ingest Linux host logs via **file monitoring inputs**
- Used SPL (Search Processing Language) to:
 - Search authentication and firewall events
 - Correlate events across multiple log sources
 - Build dashboards and alerts
- Created:
 - SSH Failed Login dashboard
 - SSH Successful Login dashboard
 - Network Reconnaissance (Nmap/UFW) dashboard

Ubuntu Linux (Monitored Host)

- Target system being monitored
- Generated authentication, firewall, kernel, and system logs
- Acted as the **victim host** in the attack scenario

Kali Linux (Attacker System)

- Used to simulate attacker behavior
- Performed:
 - Network reconnaissance
 - SSH login attempts
 - SSH brute-force attempts

UFW (Uncomplicated Firewall)

- Enabled on the Ubuntu system
- Configured to:
 - Block unsolicited network traffic
 - Log blocked packets
- UFW logs were written to:
 - `/var/log/kern.log`
 - `/var/log/syslog`
- These logs were ingested into Splunk and used to:
 - Detect Nmap port scanning
 - Identify suspicious network probing activity

Nmap

- Used from Kali Linux to perform **network reconnaissance**
- Generated firewall block events when scanning closed ports
- Activity detected via UFW firewall logs in Splunk

SSH (Secure Shell)

- Primary access service on the Ubuntu system
- Generated authentication events:
 - Failed login attempts
 - Successful login events
- Logs analyzed to:
 - Detect brute-force behavior
 - Identify successful access following failures

Hydra

- Used to simulate **SSH brute-force attacks**
- Generated multiple failed authentication attempts
- Final successful login event captured in authentication logs
- Helped demonstrate:
 - Credential attack patterns
 - Transition from failed attempts to successful access

Log Sources Ingested into Splunk

♦ `/var/log/auth.log`

- SSH authentication attempts

- Failed password attempts
- Successful SSH logins
- Used for:
 - Brute-force detection
 - Access monitoring

♦ **/var/log/kern.log**

- Kernel-level events
- UFW firewall block messages
- Network packet drop logs
- Used for:
 - Port scan detection
 - Network attack visibility

♦ **/var/log/syslog**

- General system and service-level logs
- Supplemented firewall and kernel events
- Used for:
 - Context during investigations
 - System activity correlation

Alerts Implemented

- SSH failed login alert
- SSH successful login alert

All Alerts were configured as **scheduled alerts** and tested using real attack activity.

FLOWCHART

